

- Q.20 Explain the concept of flow rate in hydraulic system. (CO6)
- Q.21 What are the advantages and disadvantages of using air as a fluid in pneumatic systems? (CO7)
- Q.22 Describe the working of a simple hydraulic circuit. (CO8)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Discuss the role of hydraulic pumps in a hydraulic system and their types. (CO4)
- Q.24 Compare the efficiency and safety features of hydraulic and pneumatic systems. (CO8)
- Q.25 Explain the construction and working of a hydraulic press, including its applications. (CO9)

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3rd Year / Advance Diploma in Tool & Die Making Subject : Hydraulics & Pneumatics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is a characteristic of hydraulic fluids?
- a) Low density b) High compressibility
c) Non-corrosive d) Low viscosity
- Q.2 A pneumatic system typically uses:
- a) Oil b) Water
c) Compressed Air d) Gasoline
- Q.3 The primary function of a hydraulic actuator is to :
- a) Control fluid flow
b) Store energy
c) Convert hydraulic energy into mechanical motion
d) Measure pressure

- Q.4 The term "fluid power" refers to the use of :
- Electricity
 - Hydraulics and pneumatics
 - Mechanical energy
 - Chemical reactions
- Q.5 The component that reduces pressure in hydraulic system is called a :
- Pump
 - Accumulator
 - Pressure relief valve
 - Filter
- Q.6 Which of the following components is essential for a pneumatic system to function?
- Hydraulic fluid
 - Pneumatic cylinder
 - Oil reservoir
 - Flow meter

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The force exerted by a fluid in a hydraulic system can be calculated using the formula _____. (CO3)
- Q.8 The device that provides the initial pressure in a hydraulic system is called a _____. (CO6)
- Q.9 _____ is the process of converting hydraulic energy into mechanical work. (CO4)

- Q.10 The major advantage of using pneumatic systems is that they are _____. (CO5)
- Q.11 In a hydraulic circuit, the pressure is measured in _____. (CO8)
- Q.12 A _____ is used to connect different components in a hydraulic system. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe the main components of a hydraulic system. (CO6)
- Q.14 What is the importance of maintaining proper fluid viscosity in hydraulic systems? (CO2)
- Q.15 How does a hydraulic motor function? (CO4)
- Q.16 Explain the working principle of a pneumatic cylinder. (CO5)
- Q.17 What are the common applications of hydraulic system in industry? (CO9)
- Q.18 Describe the function of a filter in a hydraulic system.
- Q.19 What is the purpose of using seals in hydraulic components? (CO7)